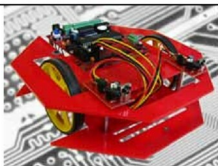


iBot

Features

- Versatile and feature-rich programmable robot development platform
- Based on the popular 8051 family microcontroller
- User programmable and can be used as a simple line follower, obstacle avoider, sumo robot and much more
- Ideal for use in robotic labs, embedded systems training, engineering projects as well as hobby robotic projects
- Easy to use for beginners and complex enough for advanced users



are available with it or which can be interfaced with it.

4. When going for a remote-controlled robot, check the kind of remote control available with it, range of the remote control and number of controls available. Also check whether the remote control has analogue input option to control the speed.

5. Check the power source of the robot. Preferably, buy a robot with rechargeable battery.

6. If going for an advanced kit, check for expandability and flexibility. Choose a kit which allows you to use the same kit for future requirements. Also check the wireless/wired connectivity with other robots and computers.

But that's not all

Most of the robotic kit vendors in India are sellers or dealers, and not manufacturers. Many of them don't know what exactly they are selling. They do not know how to use the kit and may give wrong information while selling the kits.

"Buy genuine and branded stuff. Many companies are now selling cheap items like white gear box motors which have plastic or thin metallic gear. You might save some amount in the motor but your project will suffer as it's not up to the standard," cautions Ajit Vira, partner, Vigarobokit.

Look for kits that come with a well documented manual or training programme. There are a lot of colourful kits in the market. Also look for scalability in terms of add-ons so that you can keep on developing more and more complex projects. Make sure the kit is multipurpose and reprogrammable. A robot with only one or two functions may end up like a toy after buying.

Specifications play a very important role. Before buying any kit, check its detailed specifications and compare the features with other kits. When ordering online, often people buy items based on the images provided on the website. These are usually very close-up shots and can mislead you about the size. For example, the images of humanoid robots make us feel that the robot is of the same size as humans. So don't forget to check the dimensions of the kit.

Next, check the ease of programming. Most of the robotic kits with controller need a separate programmer to load the user programs. Only higher-end kits come with a programmer. Often people buy kits without knowing this feature. Also check whether the software is user-friendly.

The kit may have the option to connect various sensors. But many providers won't supply these sensors along with the kit. They have to be

bought with extra payment. So make sure that the components are supplied with the kit.

Price varies with functionality

In general, pricing starts from a few hundred rupees and may go up to Rs 100,000. Some basic robot kits which are imported from China cost less than Rs 200. However, advanced kits like humanoids and autonomous quadrotor aircrafts cost more than Rs 50,000.

"The price basically depends upon the level and functionality of the robot. For example, a hexapod robot has 12 or 18 degrees of freedom and hence 12 or 18 motors. All these motors should be controlled simultaneously and individually. The motors, controller hardware, connectivity, remote control, software and mechanical hardware required to do this add to the total cost, making the robot costly," says Sandeep Vaidya, R&D head, Robokits India.

The price also depends upon the quality of the robot kit. For example, if a robot is made with simple toy motors, each motor will cost around Rs 120. If the same robot is made from industrial-quality motors, the motor will cost more than Rs 1500 each. And if very high-quality, efficient motors from companies like Maxon, Faulhaber or Portescap are used, each motor will cost more than Rs 5000.

The price depends on each part like mechanical structure, circuit and processor, and there are many options available which can affect the total price.

Last but not the least

Ideally, a hobby kit should be a microcontroller development board with schematic, standard sensors, reliable wireless modules, DC motors with encoders and a USB interface. The development board will help you to interface many different kinds of peripherals. ●

The author is executive editor at EFM